MCQ solutions

1. **C.** Three independent positions with four choices each, including zero at the start: $4^3 = 64$.
2. **B.** $5 \times 4 = 20$. The order of the roles matters and the same person cannot hold both.
3. **D.** $4! = 4 \times 3 \times 2 \times 1 = 24$.
4. **A.** Each element is counted once. Without disjointness, the intersection would have to be subtracted.
5. **C.** A list keeps the order and allows a card to be reused; there are $2^2 = 4$ outcomes.
6. **B.** The group is not ordered, so $\binom{8}{3} = 56$; the ordered product counts each group $3!$ times.
7. **D.** For each element, two decisions: included or not. We get $2^5 = 32$, including the empty subset.
8. **A.** Separate the groups containing a fixed element (choose two from five) from those not containing it (three out of five).
9. **C.** $\binom{5}{2} - \binom{2}{2} = 10 - 1 = 9$. Another calculation: $3 \times 2 + \binom{3}{2} = 9$. Six counts only the mixed pairs.
10. **B.** Choosing the three members of group 1 determines group 2: $\binom{6}{3} = 20$. Dividing by 2 would be correct only for unnumbered groups.

True/false. False: there is an empty subset, so $\binom{n}{0} = 1$. True: question 9. False: a simultaneous draw gives a subset; adding an order changes the object being counted.

Solutions to the applications and guidance

Application 1. Codes: 01,02,10,12,20,21. Only the last four represent numbers with two digits. A code may start with zero.

Application 2. Two red items: one outfit. Exactly one: $1 \times 2 + 3 \times 1 = 5$ outfits. The two cases are disjoint.

Application 3. Exactly two Terminale students: $\binom{6}{2}\binom{4}{1} = 60$. Mixed group: $\binom{10}{3} - \binom{6}{3} - \binom{4}{3} = 96$; or $\binom{6}{2}\binom{4}{1} + \binom{6}{1}\binom{4}{2} = 60 + 36 = 96$.

Application 4. Each unordered pair is counted twice; there is $\binom{5}{2} = 10$ pairs. For a main member and a substitute, the roles distinguish the choices, hence 20.

Application 5. Row: 1, 5, 10, 10, 5, 1, sum 32. The first 10 counts the subsets of size 2 of a set of size 5; the second those of size 3.

Application 6. Two zeros: $\binom{4}{2}4^2 = 96$. Of five digits, three are even and two odd. Sum odd: $\binom{4}{1}2\,3^3 + \binom{4}{3}2^3\,3 = 216 + 96 = 312$.

Institutionalization

Ask before each calculation: "What is a possible result? When are two results different?" Allow lists and trees in the small case, then have students make explicit what generalises. The parity calculation can remain a supported challenge; the case-by-case method is enough.

Teaching notes and solutions

Guidance for the teacher

The common test S28 comes before the combinations in S30: include only principles and lists y-axis. Written test 3 may use combinations, but not Pascal's rule taught in S35. Have students state what is being counted before any formula. For struggling students, use cards with names; does a permutation of cards change the object? The binomial theorem, mentioned in the progression, is an extension: keep it explicitly optional.

Solutions to the activity

1. A pair is a subset of two pupils among six: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

2. The president can be chosen in six ways, then the secretary from among the other five students: $6 \times 5 = 30$ distributions.

3. Each task may be given to any of the six pupils, including a pupil already chosen. The three successive choices give $6^3 = 216$ sequences.

4. With four cards, the pairs are AB, AC, AD, BC, BD, CD : six choices. The twelve pairs of roles are:

$$\begin{array}{lll} (A,B) & (A,C) & (A,D) \\ (B,A) & (B,C) & (B,D) \\ (C,A) & (C,B) & (C,D) \\ (D,A) & (D,B) & (D,C) \end{array}$$

   The number of sequences of three interventions is $4^3 = 64$, since each position has four choice.

5. For a pair, order does not matter and no pupil is repeated. For roles, order matters, without repetition. For the interventions, order matters and repetition is allowed.

E1. There are 10^4 codes in total. Those containing no 0 have nine choices for each of the four positions, i.e. 9^4 codes. By the complement, the required number is $10^4 - 9^4 = 3439$.

E2. There are ten choices for gold, then nine for silver and eight for bronze. The multiplication principle gives $10 \times 9 \times 8 = 720$ podiums. Changing the order changes the podium.

E3. The required person is already chosen. It remains to choose a subset of two people among the six others: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

E4. A path is determined by the positions of its four steps to the right among seven positions. The three other steps are then forced. We therefore count $\binom{7}{4} = 35$ paths.

E5. Choose two girls out of eight, then two boys out of seven. Each pair of choices gives only one group: $\binom{8}{2}\binom{7}{2} = 28 \times 21 = 588$.

Solutions to homework DM9

1. Order does not matter: there are $\binom{12}{4} = \frac{12\times11\times10\times9}{24} = 495$ committees.
2. Two pupils are chosen from seven and two teachers from five: $\binom{7}{2}\binom{5}{2} = 21 \times 10 = 210$ committees.
3. From the 495 committees we remove the $\binom{5}{4} = 5$ committees made only of teachers and the $\binom{7}{4} = 35$ committees made up only of students. These categories are disjoint: $495 - 5 - 35 = 455$.
4. Each of the 210 committees has two choices of teacher president and two choices of student secretary: $210 \times 2 \times 2 = 840$ arrangements.
5. Let us classify the committees by the number k of teachers, with $0 \leq k \leq 4$. The category k contains $\binom{5}{k}\binom{7}{4-k}$ committees. The five categories are disjoint and cover all the committees. By addition,
$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4} = 495.$$

Solutions to the textbook or past-paper work

A hand contains five distinct cards; order does not matter. There are therefore $\binom{52}{5} = 2{,}598{,}960$ hands. For a hand containing the ace of spades, this card is already chosen; the other four are chosen from the remaining 51 cards. There are $\binom{51}{4} = 249{,}900$ hands. This result is not multiplied by 5: the hand has no first or last position.